Interface Queue<E>

Type Parameters:

`E - the type of elements held in this collection`

All Superinterfaces:

`Collection<E>, Iterable<E>`

All Known Subinterfaces:

`BlockingDeque<E>, BlockingQueue<E>, Deque<E>, TransferQueue<E>`

All Known Implementing Classes:

`AbstractQueue, ArrayBlockingQueue, ArrayDeque, ConcurrentLinkedDeque, ConcurrentLinkedQueue, DelayQueue, LinkedBlockingDeque`

```
public interface Queue<E>
extends Collection<E>
```

A collection designed for holding elements prior to processing. Besides basic `Collection` operations, queues provide additional insertion, extraction, and inspection operations. Each of these methods exists in two forms: one throws an exception if the operation fails, the other returns a special value (either `null` or `false`, depending on the operation). The latter form of the insert operation is designed specifically for use with capacity-restricted `Queue` implementations; in most implementations, insert operations cannot fail.

Summary of Queue methods

	Throws exception	*Returns special value*
Insert	`add(e)`	`offer(e)`
Remove	`remove()`	`poll()`
Examine	`element()`	`peek()`

Queues typically, but do not necessarily, order elements in a FIFO (first-in-first-out) manner. Among the exceptions are priority queues, which order elements according to a supplied comparator, or the elements' natural ordering, and LIFO queues (or stacks) which order the elements LIFO (last-in-first-out). Whatever the ordering used, the *head* of the queue is that element which would be removed by a call to `remove()` or `poll()`. In a FIFO queue, all new elements are inserted at the *tail* of the queue. Other kinds of queues may use different placement rules. Every `Queue` implementation must specify its ordering properties.

The `offer` method inserts an element if possible, otherwise returning `false`. This differs from the `Collection.add` method, which can fail to add an element only by throwing an unchecked exception. The `offer` method is designed for use when failure is a normal, rather than exceptional occurrence, for example, in fixed-capacity (or "bounded") queues.

The `remove()` and `poll()` methods remove and return the head of the queue. Exactly which element is removed from the queue is a function of the queue's ordering policy, which differs from implementation to implementation. The `remove()` and `poll()` methods differ only in their behavior when the queue is empty: the `remove()` method throws an exception, while the `poll()` method returns `null`.

The `element()` and `peek()` methods return, but do not remove, the head of the queue.

The `Queue` interface does not define the *blocking queue methods*, which are common in concurrent programming. These methods, which wait for elements to appear or for space to become available, are defined in the `BlockingQueue` interface, which extends this interface.

`Queue` implementations generally do not allow insertion of `null` elements, although some implementations, such as `LinkedList`, do not prohibit insertion of `null`. Even in the implementations that permit it, `null` should not be inserted into a `Queue`, as `null` is also used as a special return value by the `poll` method to indicate that the queue contains no elements.

`Queue` implementations generally do not define element-based versions of methods `equals` and `hashCode` but instead inherit the identity based versions from class `Object`, because element-based equality is not always well-defined for queues with the same elements but different ordering properties.

This interface is a member of the Java Collections Framework.

Since:

`1.5`

See Also:

`Collection, LinkedList, PriorityQueue, LinkedBlockingQueue, BlockingQueue, ArrayBlockingQueue, LinkedBlockingQueue, PriorityB`

Method Summary

All Methods Instance Methods Abstract Methods

Modifier and Type	Method and Description
`boolean`	**`add`**`(E e)` Inserts the specified element into this queue if it is possible to do so immediately without violating capacity restrictions, returning `true` upon success and throwing an `IllegalStateException` if no space is currently available.
`E`	**`element`**`()` Retrieves, but does not remove, the head of this queue.
`boolean`	**`offer`**`(E e)`

	Inserts the specified element into this queue if it is possible to do so immediately without violating capacity restrictions.
E	**peek**() Retrieves, but does not remove, the head of this queue, or returns `null` if this queue is empty.
E	**poll**() Retrieves and removes the head of this queue, or returns `null` if this queue is empty.
E	**remove**() Retrieves and removes the head of this queue.

Methods inherited from interface java.util.Collection

addAll, clear, contains, containsAll, equals, hashCode, isEmpty, iterator, parallelStream, remove, removeAll, removeIf, re

Methods inherited from interface java.lang.Iterable

forEach

Method Detail

add

boolean add(E e)

Inserts the specified element into this queue if it is possible to do so immediately without violating capacity restrictions, returning `true` upon success and throwing an `IllegalStateException` if no space is currently available.

Specified by:

add in interface Collection<E>

Parameters:

e - the element to add

Returns:

true (as specified by Collection.add(E))

Throws:

IllegalStateException - if the element cannot be added at this time due to capacity restrictions

ClassCastException - if the class of the specified element prevents it from being added to this queue

NullPointerException - if the specified element is null and this queue does not permit null elements

IllegalArgumentException - if some property of this element prevents it from being added to this queue

offer

boolean offer(E e)

Inserts the specified element into this queue if it is possible to do so immediately without violating capacity restrictions. When using a capacity-restricted queue, this method is generally preferable to add(E), which can fail to insert an element only by throwing an exception.

Parameters:

e - the element to add

Returns:

true if the element was added to this queue, else false

Throws:

ClassCastException - if the class of the specified element prevents it from being added to this queue

NullPointerException - if the specified element is null and this queue does not permit null elements

IllegalArgumentException - if some property of this element prevents it from being added to this queue

remove

E remove()

Retrieves and removes the head of this queue. This method differs from poll only in that it throws an exception if this queue is empty.

Returns:

the head of this queue

Throws:

NoSuchElementException - if this queue is empty

poll

`E poll()`

Retrieves and removes the head of this queue, or returns `null` if this queue is empty.

Returns:

`the head of this queue, or null if this queue is empty`

element

`E element()`

Retrieves, but does not remove, the head of this queue. This method differs from `peek` only in that it throws an exception if this queue is empty.

Returns:

`the head of this queue`

Throws:

`NoSuchElementException - if this queue is empty`

peek

`E peek()`

Retrieves, but does not remove, the head of this queue, or returns `null` if this queue is empty.

Returns:

`the head of this queue, or null if this queue is empty`